# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING**

**COURSE OUTCOME COVERED**

***CO1:*** *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

*Assessment Tool Weightage: 10%*

**QUESTIONS: 15 Total Marks: 25** Annexure A

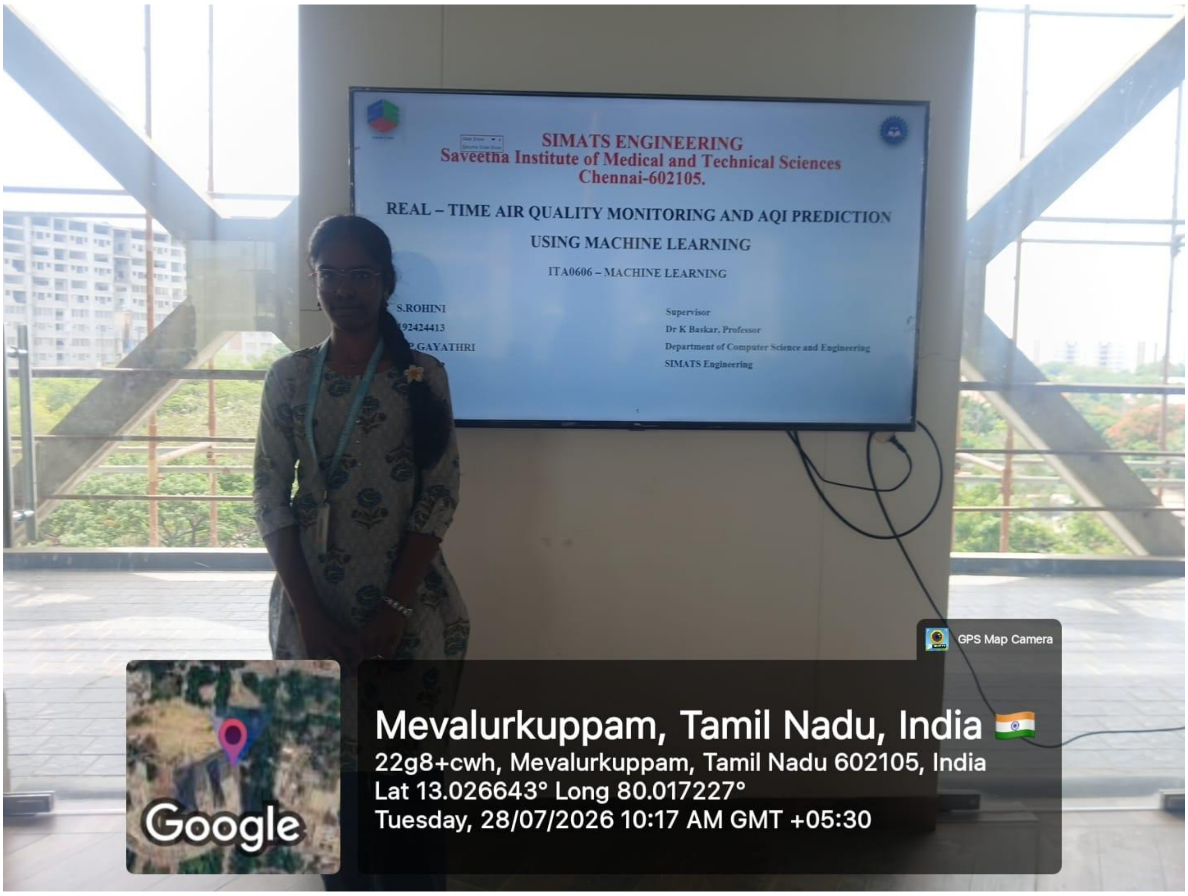
**TECHNICAL PRESENTATION**

***Code of Conduct:***

*I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: S.Rohini***

# GEOTAG PHOTO



# SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
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## REAL – TIME AIR QUALITY MONITORING AND AQI PREDICTION USING MACHINE LEARNING

**ITA0606 – MACHINE LEARNING**


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## ABSTRACT


- The objective of this project is to develop a web-based system for **Real-time air quality monitoring and Air Quality Index (AQI) prediction** using machine learning. The system collects live environmental data, including PM2.5, PM10, CO, $NO_2$, $SO_2$, $O_3$, temperature, humidity, and wind speed, through an Air Quality API.
- The collected data is preprocessed and used as input to a **Random Forest Regression** model trained on historical air quality data to predict the AQI.
- The predicted AQI is classified into standard air quality categories, and appropriate health recommendations are provided to users.
- The expected result is an interactive Streamlit web application that accurately predicts AQI, displays real-time pollutant information, visualizes air quality trends, and helps users make informed decisions to reduce health risks.



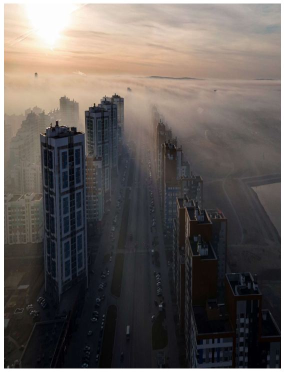

## Problem Statement

- Current air quality monitoring systems primarily display real-time pollutant values without accurately predicting future Air Quality Index (AQI), limiting proactive environmental decision-making.
- Rapid changes in pollution levels due to traffic, industrial emissions, and weather conditions make it difficult for users to assess upcoming air quality and potential health risks.

**This results in limited air quality awareness, delayed preventive actions, and increased health risks, especially for children, elderly people, and individuals with respiratory diseases.**

**Where Does the Problem Occur?**

- Urban areas with high traffic congestion and industrial activities.
- Regions experiencing rapid changes in weather and pollution levels.
- Locations where continuous air quality monitoring and early AQI prediction are required.





## Objectives

🎯 **To develop** a web-based application that collects, monitors, and displays real-time air quality data, including PM2.5, PM10, CO, $NO_2$, $SO_2$, $O_3$, temperature, humidity, and wind speed, using an Air Quality API.

🎯 **To predict** the Air Quality Index (AQI) using the Random Forest Regression algorithm, classify air quality into standard AQI categories, and provide appropriate health recommendations through an interactive dashboard.

# Literature Survey

| Sl. No | Year of Publication | Authors | Title of the Publication | Findings | Limitations |
|---|---|---|---|---|---|
| 1 | **2024** | Mulomba Mukendi Christian, Hyebong Choi | Air Quality Forecasting Using Machine Learning: A Global Perspective with Relevance to Low-Resource Settings | Random Forest provided reliable AQI prediction, especially for classification tasks in low-resource settings. | Performance depends on the availability and quality of environmental data. |
| 2 | **2024** | Kamaljeet Kaur Sidhu et al. | Predictive Modelling of Air Quality Index (AQI) Across Diverse Cities and States of India Using Machine Learning | Random Forest outperformed several ML models for AQI prediction using CPCB data. | Model focused on selected regions and required continuous data updates. |
| 3 | **2025** | Yıldırım Özüpak, Feyyaz Alpsalaz, Emrah Aslan | Air Quality Forecasting Using Machine Learning: Comparative Analysis and Ensemble Strategies for Enhanced Prediction | Ensemble machine learning techniques improved AQI forecasting accuracy. | Increased computational complexity and training time. |
| 4 | **2025** | Emine Cengil | The Power of Machine Learning Methods and PSO in Air Quality Prediction | Random Forest and optimized ML methods achieved high prediction performance. | Optimization increased model complexity and computational cost. |
| 5 | **2026** | Xiaofeng Zhu et al. | Applicability Analysis of Tree-Based Ensemble Learning for Air Pollutant Prediction Models | Random Forest achieved excellent prediction accuracy for $PM_{2.5}$ and $PM_{10}$ and offered interpretable results. | Study was validated using data from specific cities and may require adaptation for other regions. |

# System Architecture / Workflow